

a launch event con-
tributes to all
three domains

Terrestrial domain
(*T*)

raw materials
manufacturing
integration
ground operations /
EOL

EF 3.1 categories

$\tilde{I}_T$

Atmospheric domain
(*A*)

launch ascent
re-entry demise
stratospheric species
NO_x, Al₂O₃, BC, H₂O,
HCl

$\tilde{I}_A$

Orbital domain (*O*)

debris generation
(ECOB)
operational congestion
in-orbit material critical-
ity

DGP + CC + MC

$\tilde{I}_O$

**Space Sustainability
Composite Indicator**

$$SSCI_m = \mathcal{F}(\tilde{I}_T^m, \tilde{I}_A^m, \tilde{I}_O^m)$$

$$\text{linear } \sum_d w_d \tilde{I}_d \quad / \quad \text{risk-based } (\prod_d \tilde{I}_d)^{1/3}$$

dimensionless multiple of the reference-mission footprint